ME 333

MECHANICS OF MATERIALS

Lecture 5: Principal Strains & Compatibility Relations
Tutorial 1 - Concept of Strain

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Principal Components - Eigen Values

Principal Strains

- Considering the eigen value problem:

$$\det(\epsilon - \lambda I) = \begin{vmatrix} \epsilon_{xx}-\lambda & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy}-\lambda & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz}-\lambda \end{vmatrix} = 0$$

  Note that $\lambda = M$ (magnification factor). M_1, M_2 and M_3 are the three eigen values.
- The eigen vectors correspond to the principal directions.
- Strain Invariants:

$$\begin{aligned} \bar{I}_1 &= \mathrm{tr}(\boldsymbol{\epsilon}) = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} \\ \bar{I}_2 &= \epsilon_{xx}\epsilon_{yy} + \epsilon_{yy}\epsilon_{zz} + \epsilon_{xx}\epsilon_{zz} - \epsilon_{xy}^2 - \epsilon_{yz}^2 - \epsilon_{xz}^2 \\ \bar{I}_3 &= \det(\boldsymbol{\epsilon}) \end{aligned}$$

Principal Strains

[Example 4] Determine the principal strains and associated directions.

$$\boldsymbol{\epsilon} = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$$

- Step 1: Determine eigen values:

$$\det(\boldsymbol{\epsilon}) = \begin{vmatrix} 1-\lambda & 5 \\ 2 & 4-\lambda \end{vmatrix} = 0 \implies (\lambda+1)(\lambda-6) = 0 \implies \lambda_1 = 6, \lambda_2 = -1$$

- Step 2: Determine eigen vectors:

$$(\boldsymbol{\epsilon} - \lambda_1 \boldsymbol{I})\boldsymbol{x}_1 = \begin{bmatrix} -5 & 5 \\ 2 & -2 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix} = 0 \implies \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} \pm(1/\sqrt{2}) \\ \pm(1/\sqrt{2}) \end{Bmatrix}$$

Note : Do not forget to normalize.

Review of Basics of Tensors

$\boxed{Q}$ What is rank of a matrix?

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 1 \end{bmatrix}$$

Rank : Maximum of number of linearly independent column vectors; Rank(A) = 3

$\boxed{Q}$ For a tensor is order same as rank akin that of a matrix?

NO

Generalized Strain Description

$\boxed{Q}$ If every displacement field $\boldsymbol{\Delta}$ is not unique then are all such $\boldsymbol{\Delta}$ valid ?

Strain Compatibility Relations

Planar Small Strain Theory

- The strain - displacement relations can be written as:

$$\epsilon_{xx} = \frac{\partial u}{\partial x}; \quad \epsilon_{yy} = \frac{\partial v}{\partial y}; \quad \epsilon_{xy} = \frac{1}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)$$

- Considering strains in $x - y$ plane, differentiating as below :

$$\frac{\partial^2 \epsilon_{xx}}{\partial y^2} = \frac{\partial^3 u}{\partial x \partial y^2}; \quad \frac{\partial^2 \epsilon_{yy}}{\partial x^2} = \frac{\partial^3 v}{\partial x^2 \partial y}; \quad 2\frac{\partial^2 \epsilon_{xy}}{\partial x \partial y} = \frac{\partial^3 u}{\partial x \partial y^2} + \frac{\partial^3 v}{\partial x^2 \partial y}$$

- The relation between the normal and shear strain components is:

$$\frac{\partial^2 \epsilon_{xx}}{\partial y^2} + \frac{\partial^2 \epsilon_{yy}}{\partial x^2} = 2\frac{\partial^2 \epsilon_{xy}}{\partial x \partial y}$$

Strain Compatibility Relations

- In a full three dimensional setting:

$$\begin{aligned}
\epsilon_{xx,yy} + \epsilon_{yy,xx} &= 2\epsilon_{xy,xy} \\
\epsilon_{yy,zz} + \epsilon_{zz,yy} &= 2\epsilon_{yz,yz} \\
\epsilon_{zz,xx} + \epsilon_{xx,zz} &= 2\epsilon_{xz,xz} \\
\epsilon_{zz,xy} + \epsilon_{xy,zz} &= \epsilon_{yz,zx} + \epsilon_{zx,yz} \\
\epsilon_{yy,xz} + \epsilon_{xz,yy} &= \epsilon_{yz,yx} + \epsilon_{xy,yz} \\
\epsilon_{xx,yz} + \epsilon_{yz,xx} &= \epsilon_{xz,xy} + \epsilon_{xy,xz}
\end{aligned}$$

- These six relations are known as *Compatibility Relations*
- Note that in the above set of equations the $(,)$ is used to denote partial differentiation with respect to the subscripts following the comma.

Strain Compatibility Relations

[Example 1] Is the given plane strain state permissible:

$$\epsilon_{xx} = k(x^2 + y^2); \quad \epsilon_{yy} = k(y^2 + x^2); \quad \epsilon_{xy} = k'xyz$$

where k and k' are small constants.

In the plane strain state $\epsilon_{zz} = \epsilon_{xz} = \epsilon_{yz} = 0$ and therefore the number of compatibility equations reduce to one.

$$\epsilon_{xx,yy} + \epsilon_{yy,xx} = 2\epsilon_{xy,xy} \implies 2k + 2k = 2k'z \implies (2k/k') = z$$

From above, the LHS is a constant while RHS is a variable and therefore, the strain state is *not* permissible.

Measurement of Strain

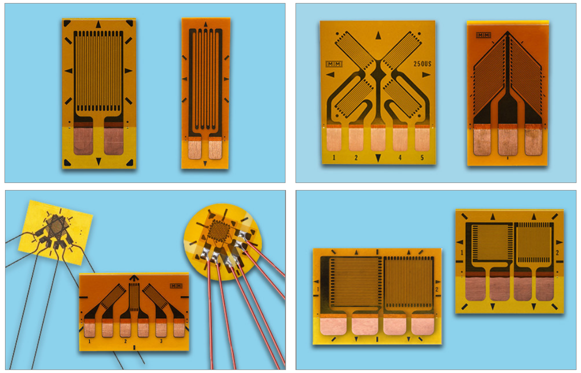
Strain Gages and Strain Rosette

Linear Patterns

Shear/Torque Rosettes

Rectangular Rosettes

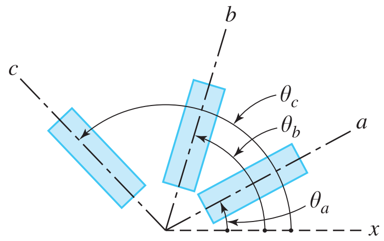
Tee Rosettes



Ref Figs: Vishay Micro-measurements

Measurement of Strain

Strain Rosette



- The strain transformation equations are:

$$\epsilon_a = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta_a + \frac{\gamma_{xy}}{2}\sin 2\theta_a$$

$$\epsilon_b = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta_b + \frac{\gamma_{xy}}{2}\sin 2\theta_b$$

$$\epsilon_c = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta_c + \frac{\gamma_{xy}}{2}\sin 2\theta_c$$

- Note that the same can be obtained using:

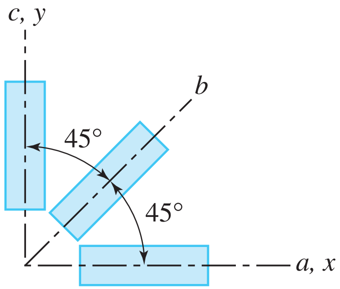
$$\epsilon_a = l_1^2\epsilon_x + m_1^2\epsilon_y + 2l_1 m_1 \epsilon_{xy}$$

$$\vdots$$

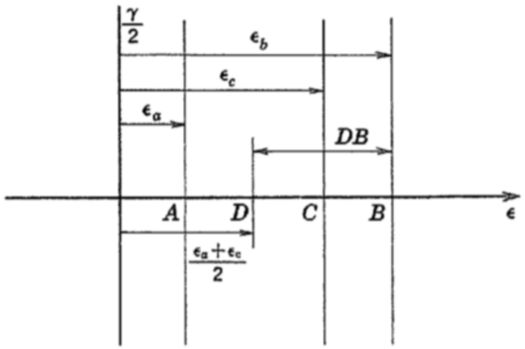
- Two configurations of strain rosette are most typically used which are: $0^\circ - 45^\circ - 90^\circ$, $0^\circ - 60^\circ - 120^\circ$ configurations.

Mohr's Circle for Strain

Strain Rosette Example : [Example 4.4]



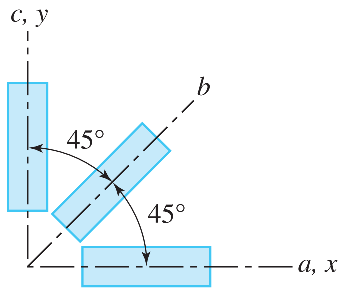
$\epsilon_a < \epsilon_c < \epsilon_b$
is assumed



Ref: An Introduction to Mechanics of Solids, Crandall et. al.

Mohr's Circle for Strain

Strain Rosette Example : [Example 4.4]



$\epsilon_a < \epsilon_c < \epsilon_b$
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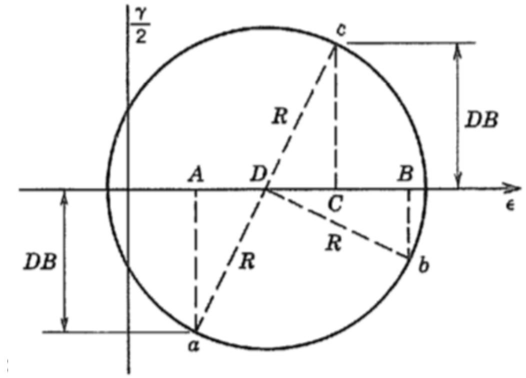
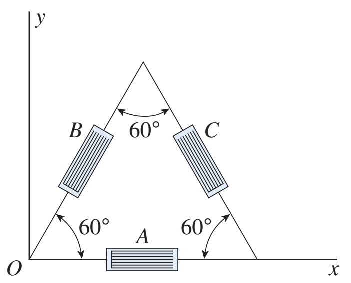


Ref: An Introduction to Mechanics of Solids, Crandall et. al.

Mohr's Circle for Strain

Strain Rosette : [Example 2] A 60° strain rosette (delta rosette) is used to measure strains ϵ_a, ϵ_b and ϵ_c, along three directions A, B and C, respectively, as shown. Obtain ϵ_{xx}, ϵ_{yy} and ϵ_{xy}.



- Step 1: Using transformation relations:

For $\theta = 0° : l_1 = 1,\ m_1 = 0 \implies \epsilon_A = \epsilon_{xx}$

For $\theta = 60° : l_1 = 1/2,\ m_1 = \sqrt{3}/2$

$$\epsilon_B = \frac{\epsilon_{xx}}{4} + \frac{3\epsilon_{yy}}{4} + \frac{\sqrt{3}\epsilon_{xy}}{2}$$

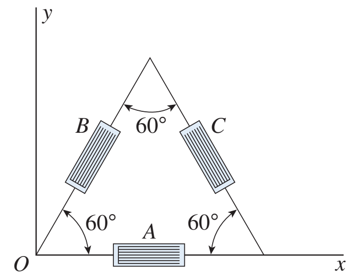
For $\theta = 120° : l_1 = -1/2,\ m_1 = \sqrt{3}/2$

$$\epsilon_C = \frac{\epsilon_{xx}}{4} + \frac{3\epsilon_{yy}}{4} - \frac{\sqrt{3}\epsilon_{xy}}{2}$$

Ref: Mechanics of Materials, James M. Gere & Barry J. Goodno

Mohr's Circle for Strain

Strain Rosette : [Example 2] (Contd ...)



- Step 2: Solve for the remaining two unknowns, we get:

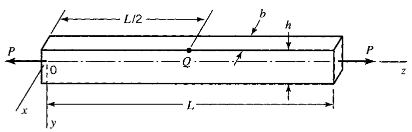
$$\epsilon_{yy} = \frac{1}{3}\left(2\epsilon_B + 2\epsilon_C - \epsilon_A\right)$$
$$\epsilon_{xy} = \frac{1}{\sqrt{3}}\left(\epsilon_B - \epsilon_C\right)$$

Ref: Mechanics of Materials, James M. Gere & Barry J. Goodno

Tutorial Problems

[Problem 2.53] The tension member shown has the following dimensions: $L = 5\,\text{m}$, $b = 100\,\text{mm}$ and $h = 200\,\text{mm}$. The (x, y, z) coordinate axes are parallel to the edges of the member, with origin O located at the centroid of the left end. Under the deformation produced by the load P, the origin O remains located at the centroid of the left end and the coordinate axes remain parallel to the edges of the deformed member. Under the action of load P, the bar elongates $20\,\text{mm}$. Assume that the volume of the bar remains constant with $\epsilon_{xx} = \epsilon_{yy}$.

1. Determine the displacements for the member and the state of strain at point Q, assuming small strain theory.
2. Determine ϵ_{zz} at point Q based on the assumption that the displacements are not small.



Tutorial Problems

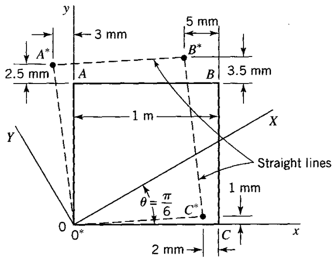
[Problem 2.53] Solution :

Answer : a) $u = -0.002x$, $v = -0.002y$ and $w = 0.004z$; b) $\epsilon_{zz} = 0.004008$ valid at all points of the body.

Tutorial Problems

[Problem 2.56] A square glass block in the side of a skyscraper is loaded so that the block is in a state of plane strain ($\epsilon_{zz} = \epsilon_{zx} = \epsilon_{zy} = 0$).

1. Determine the displacements in the block for the deformations shown and the strain components for the (x, y) coordinate axes.
2. Determine the strain components for the (X, Y) axes.



Tutorial Problems

[Problem 2.56] Solution :

Answer : a) $u = -0.002x - 0.003y$, $v = 0.001x + 0.0025y$; b) $\epsilon_{XX} = -0.00174$, $\epsilon_{YY} = 0.00224$ and $\epsilon_{XY} = 0.00145$.

Reading Assignment

Chapter 2: Sections 2.7 - 2.9